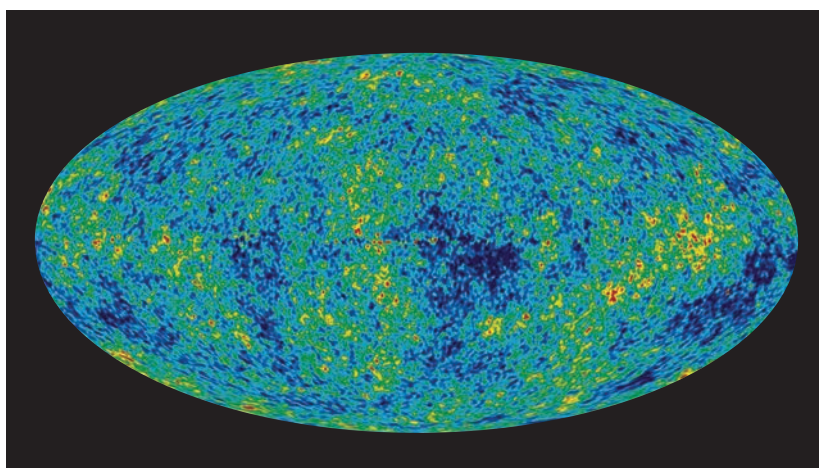


7. In 1802, spectra from stars and galaxies were first observed.

In the 1940s the Steady State Theory of the origin of the Universe was proposed by Sir Fred Hoyle and other scientists.

In 1964, American astronomers discovered there was electromagnetic interference coming from all parts of the sky at all times. This is shown in the image below.



- (a) State the name of the region of the electromagnetic spectrum that is recorded on the image shown above. [1]

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- (b) Explain the evidence that led to an alternative, different theory of the origin of the Universe to the Steady State Theory. [6 QER]

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Examiner
only

7

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THE PERIODIC TABLE

	Group
1	1
2	2
3	3
4	4
5	5
6	6
7	7
0	0

1 H Hydrogen 1																																			
7	Li	9	Be											11	B	12	C	14	N	16	O	19	F	20	Ne										
	Lithium		Beryllium												Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon										
23	Na	24	Mg											27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar										
	Sodium		Magnesium												Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon										
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Potassium		Calcium		Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	106	Rh	108	Pd	112	Ag	115	Cd	119	In	122	Sn	127	I	128	Te	131	Xe		
	Rubidium		Strontium		Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Rhodium		Silver		Cadmium		Indium		Tin		Antimony		Iodine		Tellurium		Xenon		
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	195	Ir	197	Pt	201	Au	204	Hg	207	Pb	209	Bi	210	Po	210	At	222	Rn		
	Caesium		Barium		Lanthanum		Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Mercury		Lead		Bismuth		Polonium		Astatine		Radon				
223	Fr	226	Ra	227	Ac																														
	Francium		Radium		Actinium																														

Key

relative atomic mass

Name	atomic number
7	